

Q # 1 Write short answers of the following

1. How does pipelining improve the performance of CPU?
2. Define cache hit and cache miss?
3. Define Instruction format?
4. What is direct mapping of cache?
5. What is paging in virtual memory?

			(15)
CLO2	PLO1	C2	
CLO1	PLO1	C2	
CLO1	PLO1	C1	
CLO2	PLO2	C1	
CLO2	PLO2	C2	

Q # 2 Discuss in detail

1. Describe the need of I/O interface? Also discuss the interrupts in 8085 microprocessor?
2. What is multithreading? Discuss coarse-grained and fine-grained multithreading?
3. Differentiate between out-of-order execution and speculation in instruction level parallelism?
4. Differentiate between data stripping and data mirroring in RAID 0 and RAID 1 respectively?

			(20)
CLO3	PLO2	C2	
CLO3	PLO2	C2	
CLO3	PLO2	C2	
CLO3	PLO2	C2	

Q # 3 Analyze the following

1. A computer System in using demand paging with the following reference string
7,0,1,2,0,3,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Using the FIFO algorithm show step by step how pages will be loaded and calculate the number of page faults?
(Assume three frames)

CLO2 PLO2 C2

2. A CPU has a clock cycle time of 2 ns and a five stage pipeline. Assume that each stage takes one clock cycle and there are 100 instructions to be executed

CLO2 PLO2 C2

- Calculate the total execution time with and without pipelining
- Determine the speedup ratio achieved due to pipelining

3. A cache memory has the following parameters:

CLO2 PLO2 C2

- a. Cache access time = 20ns, main memory access time = 100 ns, Hit ratio = 0.9

- b. Compare the effective memory access time (EMAT). Also, explain how increasing the cache size may effect this value.

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